LenScribe: Beamforming Subtitle Glasses

A Real-time Speech Display for Enhanced Communication


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ABSTRACT

LenScribe provides a more inclusive communication aid than traditional sound amplification. This device transcribes spoken words into text and displays them on the lens of Augmented Reality (AR) glasses for the wearer to read. The system utilizes a beamforming four microphone array and an AI-integrated speech-to-text machine (STM) to capture and process directional speech. The transcribed text will be displayed onto the AR glasses with embedded power and interface hardware.

SYSTEM OVERVIEW

Custom Printed Circuit Board (PCB):
- 4 layer board
- 5v and 3.3v power inputs
- 4-microphone array
- Audio low pass Bessel filter
- Signal amplifier for audio
- 16-bit SAR ADC
- Switched capacitor regulator

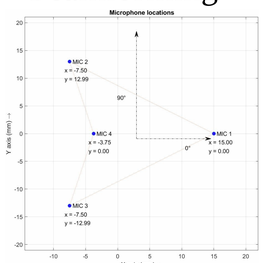
Augmented Reality Display:
- Clip-on overlay
- Lens system
- SSD1306 text display

Central Raspberry Pi Controller:
- Speech-to-Text machine
- Display driver
- System power supply
- PCB clock source

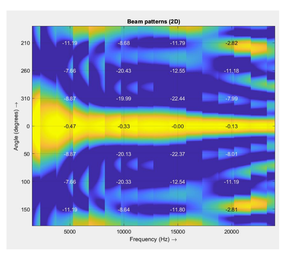
SUBSYSTEM DESIGN

Microphone Array

Array filters out surrounding noise to capture speech with more clarity and from a specific direction
- Combines signals from four microphones using Max SNR Beamforming

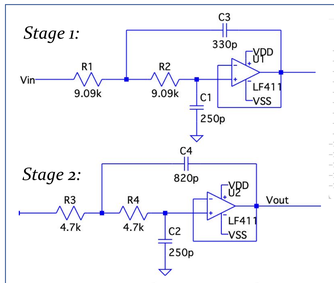


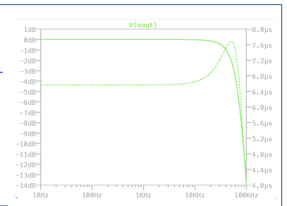
Microphone Locations



Beam Pattern Heat Map

Audio Filtering



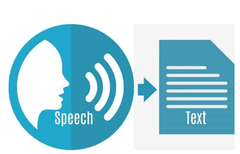


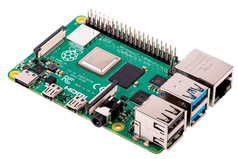
Simulated Frequency Response and Group Delay

Designed anti-aliasing low pass filter for audio intake from microphones
- 4th order Bessel Filter
  - ideal for audio due to linear phase response and minimal distortion
- Passband frequency of 8kHz
  - ideal for speech processing

Speech-to-Text

An AI-integrated speech-to-text machine was used for Automatic Speech Recognition (ASR)
- Running on Raspberry Pi
- 80k sample rate per second
- 8k frames per buffer

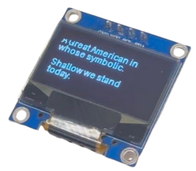




Raspberry Pi



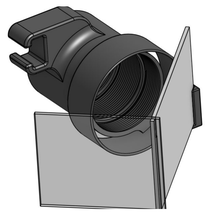
Audio input with USB microphone for testing



Text display with accurate speech-to-text

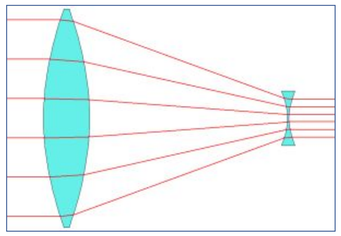
AR Display

The AR display:
- 1 inch OLED screen
- Galilean optical setup
- Mirror
- Piece of semi transparent plastic



AR Display Design

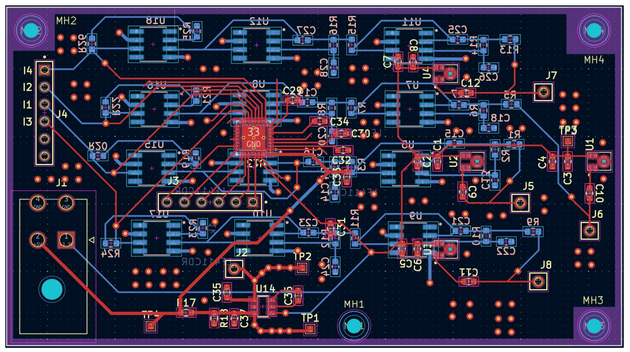
The OLED image is collimated and shrunk through the Galilean setup, and then projected onto the semi-transparent plastic.



Galilean System: optical magnifying system

INTEGRATION

PCB:

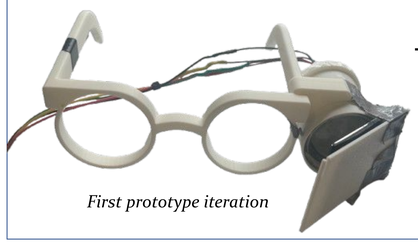


1. Audio is captured by microphone array
2. Speech is filtered and amplified
3. Converted to digital signal with ADC
4. ASR analysis on Pi
5. Text is sent to AR display to be read

PCB is integrated with the Raspberry Pi using a SPI bus.

PROTOTYPING

3D-Printed Frame:
- Custom-designed & printed frame to house components securely



First prototype iteration

- Power supply and processing housed in a wearable pack

ACKNOWLEDGEMENTS

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